

Physical Chemistry (II)

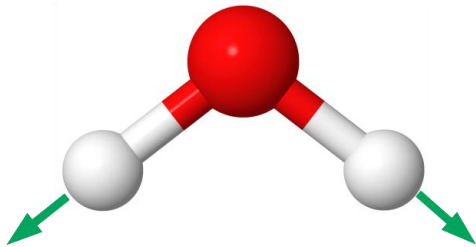
Lecture 08

Rigid Rotor

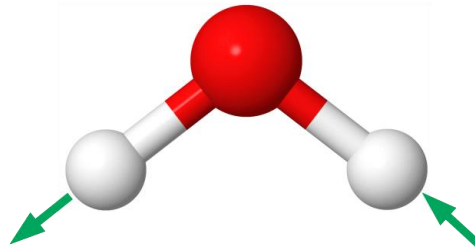


Last class

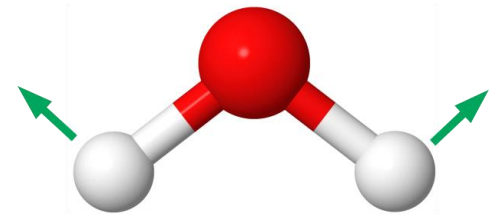
Normal Modes



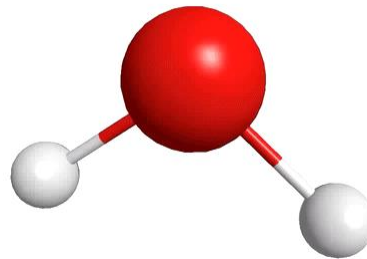
symmetric stretching



asymmetric stretching



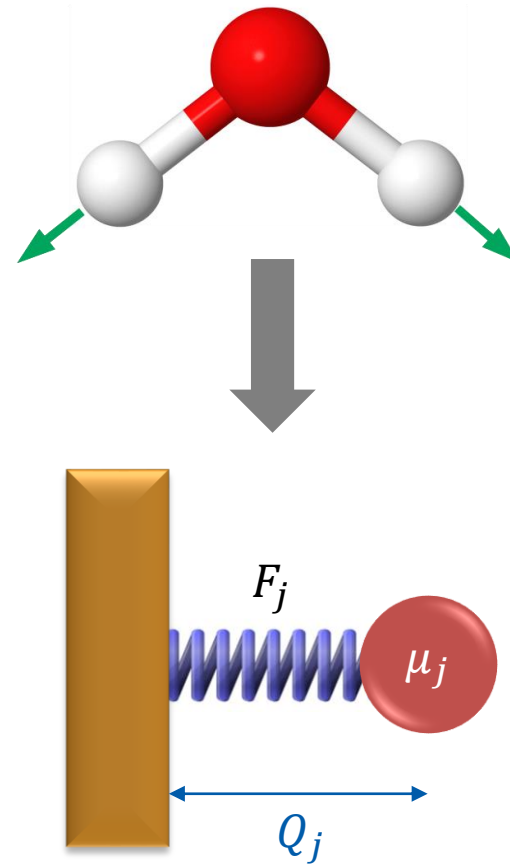
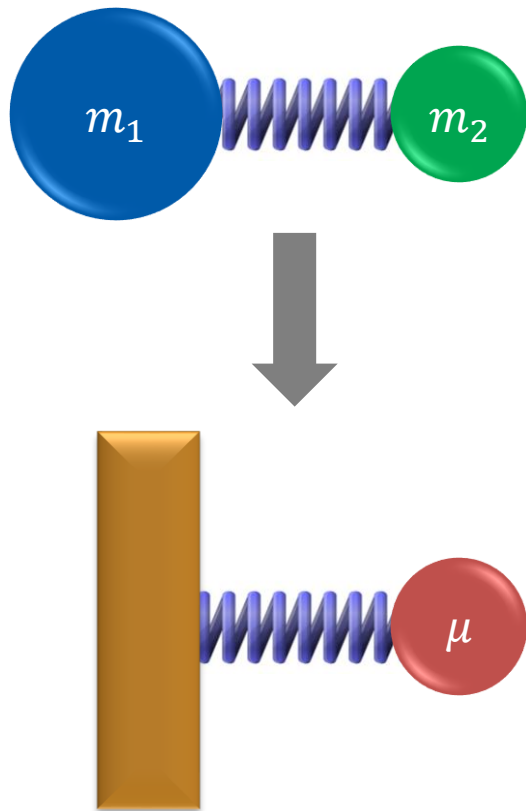
bending



(Images: https://strawberryfields.ai/phononics/_images/water.png
<https://thepaperccny.com/2022/02/28/harmony-of-the-ocean/>)

Last class

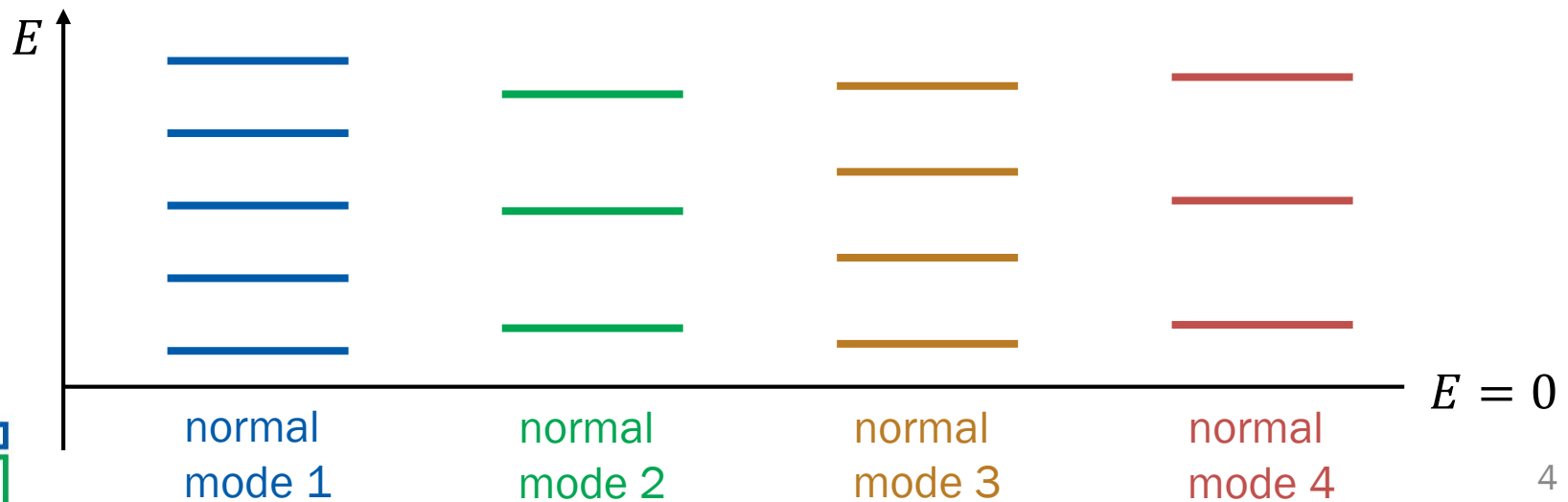
Conversion



Vibration of Polyatomics

$$E_{\text{vib}} = \sum_{j=1}^{N_{\text{vib}}} E_{\text{vib},j} = \sum_{j=1}^{N_{\text{vib}}} h\nu_j \left(v_j + \frac{1}{2} \right)$$

(where $\nu_j = \sqrt{F_j/\mu_j}$)



Selection Rules

$$\Delta v_j = \pm 1$$

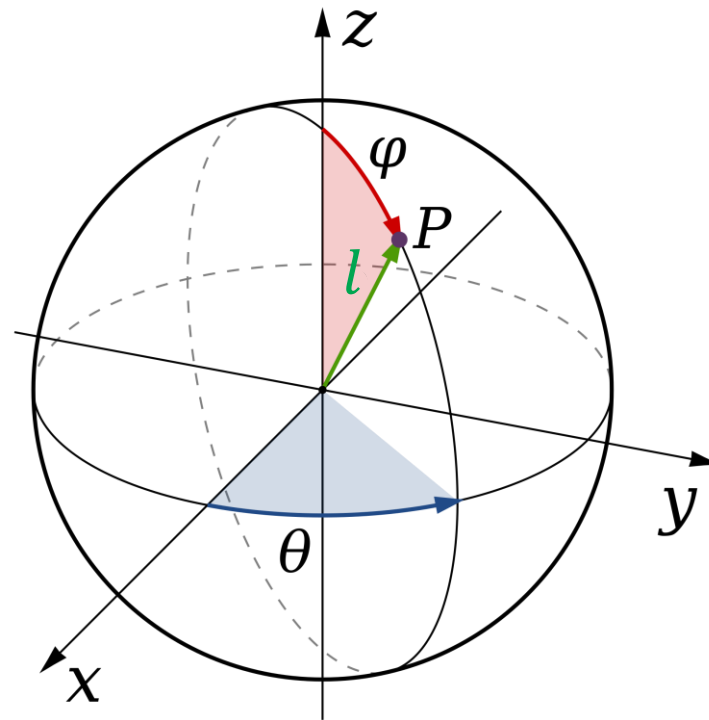
$$\frac{d\mu}{dQ_j} \neq 0$$

Dipole moment of the molecule must vary during a vibration

- **Infrared active:** the normal mode whose $d\mu/dQ_j \neq 0$
- **Infrared inactive:** the normal mode whose $d\mu/dQ_j = 0$

Last class

Rigid Rotor Model (a.k.a. Particle on a Sphere)



(Image: [https://commons.wikimedia.org/wiki/File:Spherical_Coordinates_\(Colatitude,_Longitude\).svg](https://commons.wikimedia.org/wiki/File:Spherical_Coordinates_(Colatitude,_Longitude).svg))

Last class

Angular Momenta in QM

$$\hat{L}_x = y\hat{p}_z - z\hat{p}_y = -i\hbar \left(y \frac{\partial}{\partial z} - z \frac{\partial}{\partial y} \right)$$

$$\hat{L}_y = z\hat{p}_x - x\hat{p}_z = -i\hbar \left(z \frac{\partial}{\partial x} - x \frac{\partial}{\partial z} \right)$$

$$\hat{L}_z = x\hat{p}_y - y\hat{p}_x = -i\hbar \left(x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right)$$

$$\hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2$$

$$[\hat{L}_x, \hat{L}_y] = i\hbar \hat{L}_z$$

$$[\hat{L}_y, \hat{L}_z] = i\hbar \hat{L}_x$$

$$[\hat{L}_z, \hat{L}_x] = i\hbar \hat{L}_y$$

$$[\hat{L}^2, \hat{L}_x] = 0$$

$$[\hat{L}^2, \hat{L}_y] = 0$$

$$[\hat{L}^2, \hat{L}_z] = 0$$



Last class

Hamiltonian for Rigid Rotor

$$\hat{H} = \frac{\hat{L}^2}{2I} = \frac{\hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2}{2I}$$

$$\hat{H} = -\frac{\hbar^2}{2I} \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right]$$

$$\hat{H}\psi(\theta, \phi) = E\psi(\theta, \phi)$$

$$-\frac{\hbar^2}{2I} \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right] \psi(\theta, \phi) = E\psi(\theta, \phi)$$



Let Us Solve It

$$-\frac{\hbar^2}{2I} \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right] \psi(\theta, \phi) = E \psi(\theta, \phi)$$

$$\left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right] \psi(\theta, \phi) = -\underbrace{\frac{2IE}{\hbar^2}}_{\beta} \psi(\theta, \phi)$$

Multiplying both sides by $\sin^2 \theta$,

$$\sin \theta \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \psi}{\partial \theta} \right) + \frac{\partial^2 \psi}{\partial \phi^2} + (\beta \sin^2 \theta) \psi = 0$$

Separation of Variables

$$\sin \theta \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \psi}{\partial \theta} \right) + \frac{\partial^2 \psi}{\partial \phi^2} + (\beta \sin^2 \theta) \psi = 0$$

Suppose that $\psi(\theta, \phi) = \Theta(\theta)\Phi(\phi)$ and we get

$$\frac{\sin \theta}{\Theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + \beta \sin^2 \theta = K$$

$$\frac{1}{\Phi} \frac{d^2 \Phi}{d\phi^2} = -K$$

ϕ Equation

$$\frac{1}{\Phi} \frac{d^2 \Phi}{d\phi^2} = -K$$

The boundary condition: $\Phi(\phi) = \Phi(\phi + 2\pi)$

The non-trivial solution of the ϕ equation requires

$$K = m^2 \geq 0$$

and the solution is

$$\Phi(\phi) = \frac{1}{\sqrt{2\pi}} e^{im\phi} \quad (m: \text{integer})$$

RS 4.3

M 6.6

θ Equation

$$\frac{\sin \theta}{\Theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + \beta \sin^2 \theta = K$$

$$\sin \theta \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + \beta \sin^2 \theta \Theta = m^2 \Theta$$

No simple method to solve this equation

But we “know” the solutions (again!)



Solution for the θ Equation

$$\Theta(\theta) = \left[\frac{2J+1}{2} \frac{(J-|m|)!}{(J+|m|)!} \right]^{1/2} P_J^{|m|}(\cos \theta)$$

with

$$J = 0, 1, 2, \dots \text{ and } m = 0, \pm 1, \pm 2, \dots, \pm J$$

where the associated Legendre function is

$$P_J^{|m|}(x) = \frac{(-1)^{|m|}}{2^J J!} (1-x^2)^{|m|/2} \frac{d^{J+|m|}}{dx^{J+|m|}} (x^2-1)^J$$

Associated Legendre Functions

$$P_0^0(x) = 1$$

$$P_1^0(x) = x$$

$$P_1^1(x) = (1 - x^2)^{1/2}$$

$$P_2^0(x) = \frac{1}{2}(3x^2 - 1)$$

$$P_2^1(x) = 3x(1 - x^2)^{1/2}$$

$$P_2^2(x) = 3(1 - x^2)$$

$\vdots$

Spherical Harmonics

$$\psi(\theta, \phi) = \Theta_J^{|m|}(\theta) \Phi_m(\phi)$$

The wavefunctions are called **spherical harmonics**:

$$Y_J^m(\theta, \phi) = \Theta_J^{|m|}(\theta) \Phi_m(\phi)$$

$$Y_J^m(\theta, \phi) = \underbrace{i^{m+|m|} \left[\left(\frac{2J+1}{4\pi} \right) \frac{(J-|m|)!}{(J+|m|)!} \right]^{1/2}}_{\text{convention}} P_J^{|m|}(\cos \theta) e^{im\phi}$$

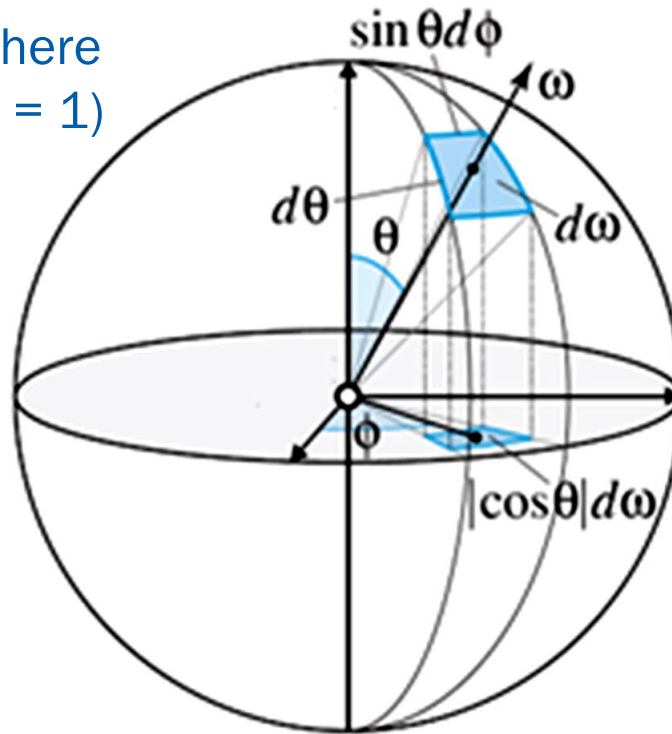
with

$$J = 0, 1, 2, \dots \text{ and } m = 0, \pm 1, \pm 2, \dots, \pm J$$

Not in text

On a Sphere

unit sphere
(radius = 1)



infinitesimal “area” = $d\omega = \sin\theta d\theta d\phi$

Integration on a Sphere

$$d\omega = \sin \theta \, d\theta \, d\phi$$

$$\int f(\theta, \phi) \, d\omega = \int_0^{2\pi} \int_0^\pi f(\theta, \phi) \sin \theta \, d\theta \, d\phi$$

- **Probability** $|Y_J^m(\theta, \phi)|^2 = [Y_J^m(\theta, \phi)]^* Y_J^m(\theta, \phi)$
- **Normalization** $\int_0^{2\pi} \int_0^\pi [Y_J^m(\theta, \phi)]^* Y_J^m(\theta, \phi) \sin \theta \, d\theta \, d\phi = 1$
- **Orthogonality** $\int_0^{2\pi} \int_0^\pi [Y_{J'}^{m'}(\theta, \phi)]^* Y_J^m(\theta, \phi) \sin \theta \, d\theta \, d\phi = 0$

Example 6-7

Show that $Y_1^{-1}(\theta, \phi)$ is normalized and that it is orthogonal to $Y_2^1(\theta, \phi)$.

$$Y_J^m(\theta, \phi) = i^{m+|m|} \left[\left(\frac{2J+1}{4\pi} \right) \frac{(J-|m|)!}{(J+|m|)!} \right]^{1/2} P_J^{|m|}(\cos \theta) e^{im\phi}$$

$$Y_1^{-1}(\theta, \phi) = i^{-1+1} \left[\left(\frac{2 \times 1 + 1}{4\pi} \right) \frac{(1-1)!}{(1+1)!} \right]^{1/2} P_1^1(\cos \theta) e^{-i\phi}$$

As $P_1^1(x) = (1-x^2)^{1/2}$,

$$Y_1^{-1}(\theta, \phi) = \left[\left(\frac{3}{4\pi} \right) \frac{1}{2} \right]^{1/2} (1 - \cos^2 \theta)^{1/2} e^{-i\phi} = \left(\frac{3}{8\pi} \right)^{1/2} \sin \theta e^{-i\phi}$$

Example 6-7

Show that $Y_1^{-1}(\theta, \phi)$ is normalized and that it is orthogonal to $Y_2^1(\theta, \phi)$.

$$Y_1^{-1}(\theta, \phi) = \left(\frac{3}{8\pi}\right)^{1/2} \sin \theta e^{-i\phi}$$

Normalization requires

$$\int_0^{2\pi} \int_0^\pi [Y_1^{-1}(\theta, \phi)]^* Y_1^{-1}(\theta, \phi) \sin \theta d\theta d\phi = 1$$

$$\begin{aligned} \text{LHS} &= \int_0^{2\pi} \int_0^\pi \left(\frac{3}{8\pi}\right)^{1/2} \sin \theta e^{i\phi} \cdot \left(\frac{3}{8\pi}\right)^{1/2} \sin \theta e^{-i\phi} \cdot \sin \theta d\theta d\phi \\ &= \frac{3}{8\pi} \int_0^\pi \sin^3 \theta d\theta \int_0^{2\pi} d\phi = \frac{3}{8\pi} \times \frac{4}{3} \times 2\pi = 1 \end{aligned}$$

Example 6-7

Show that $Y_1^{-1}(\theta, \phi)$ is normalized and that it is orthogonal to $Y_2^1(\theta, \phi)$.

$$Y_1^{-1}(\theta, \phi) = \left(\frac{3}{8\pi}\right)^{1/2} \sin \theta e^{-i\phi}$$

Similarly, one can obtain

$$Y_2^1(\theta, \phi) = -\left(\frac{15}{8\pi}\right)^{1/2} \sin \theta \cos \theta e^{i\phi}$$

$$\int_0^{2\pi} \int_0^\pi [Y_1^{-1}(\theta, \phi)]^* Y_2^1(\theta, \phi) \sin \theta d\theta d\phi = 0$$

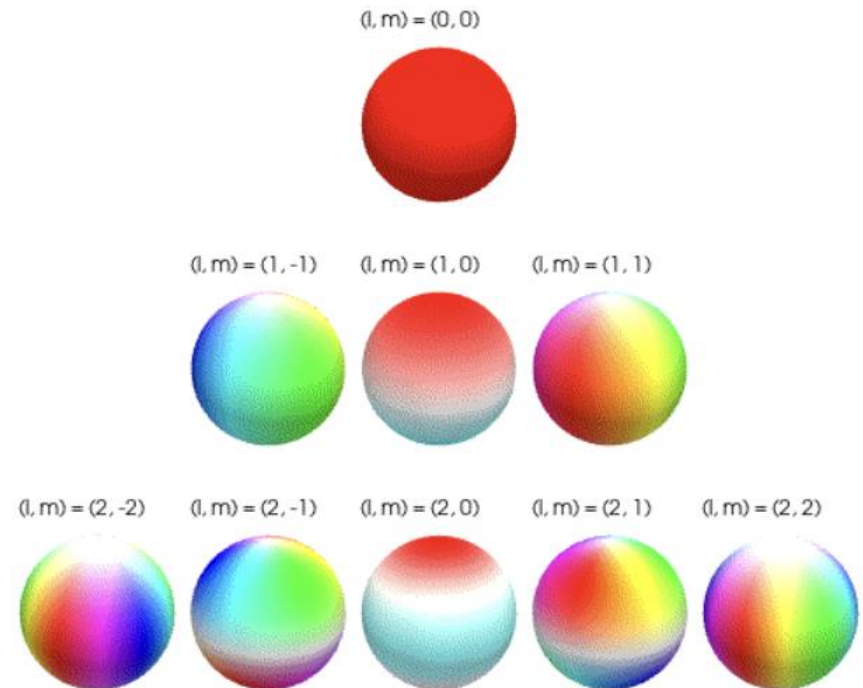
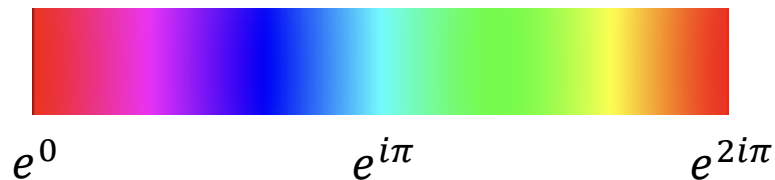
$$= -\frac{3\sqrt{5}}{8\pi} \int_0^{2\pi} \int_0^\pi \sin^3 \theta \cos \theta e^{2i\phi} d\theta d\phi = -\frac{3\sqrt{5}}{8\pi} \int_0^\pi \sin^3 \theta \cos \theta d\theta \underbrace{\int_0^{2\pi} e^{2i\phi} d\phi}_{=0} = 0$$

Not in text

“Shape” of $Y_J^m(\theta, \phi)$

$$Y_J^m(\theta, \phi) = \frac{1}{\sqrt{2\pi}} \Theta_J^{|m|}(\theta) e^{im\phi}$$

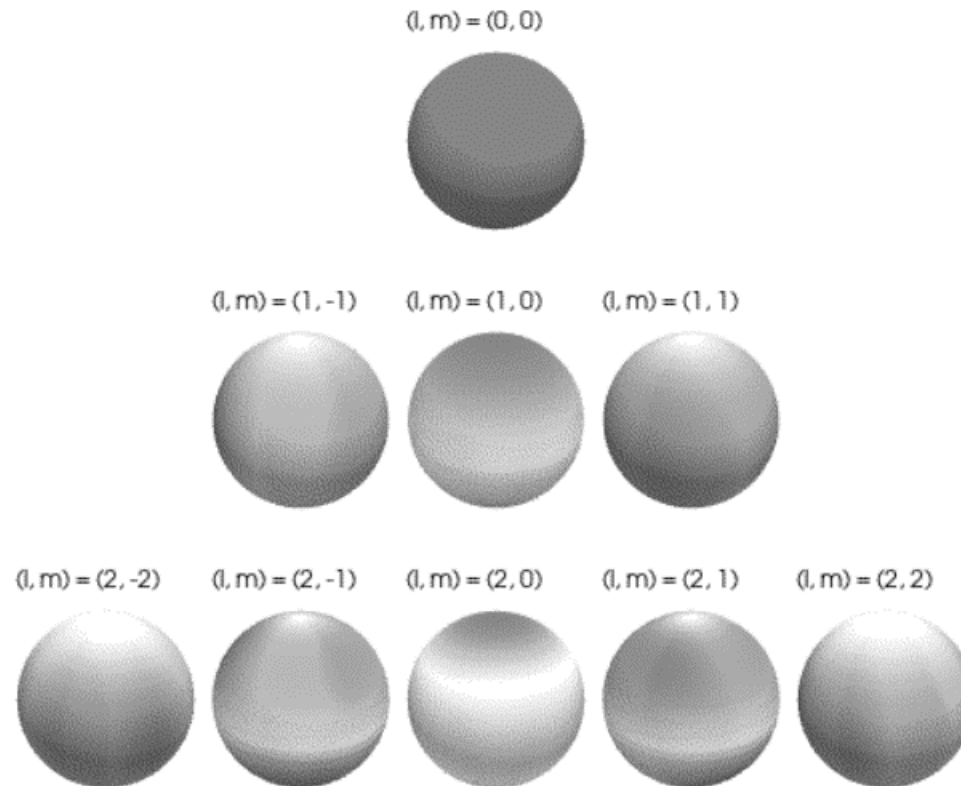
“Hue” indicates the e^{ix} part:



(Image: https://commons.wikimedia.org/wiki/File:Complex_Spherical_Harmonics_Figure_Table_Complex_Polar_Plot.gif) 21

Not in text

$$|Y_J^m(\theta, \phi)|^2$$



(Image: https://commons.wikimedia.org/wiki/File:Complex_Spherical_Harmonics_Figure_Table_Complex_Polar_Plot.gif) 22

Solution for Schrödinger Equation

- Quantum numbers: J and m , such that

$$J = 0, 1, 2, \dots \text{ and } m = 0, \pm 1, \pm 2, \dots, \pm J$$

- Wavefunction: spherical harmonics $Y_J^m(\theta, \phi)$
- Energy

$$E_{J,m} = \frac{\hbar^2}{2I} J(J+1) = \overset{\text{rotational constant } \hbar^2/2I}{\textcircled{B}} J(J+1) = E_J$$

with degeneracy $g_J = 2J + 1$

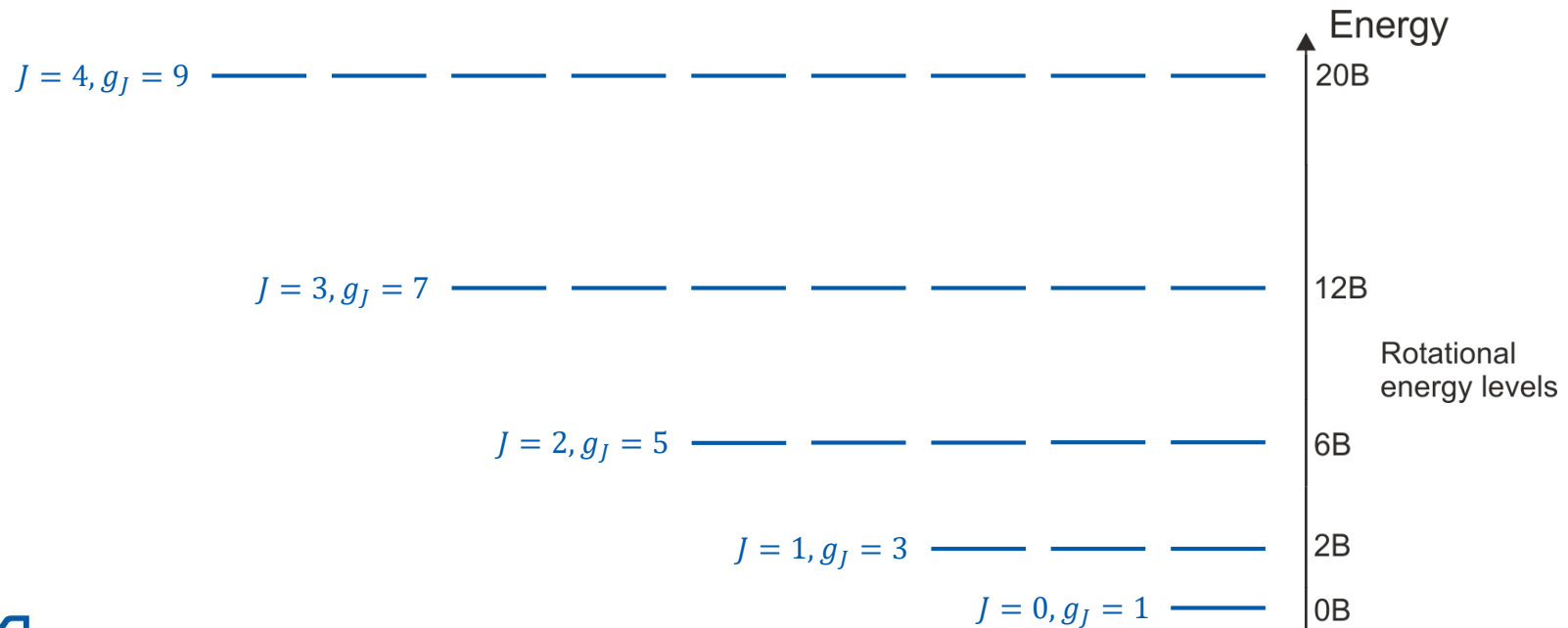
RS 4.3

M 6.1

Energy Levels

$$E_J = BJ(J + 1)$$

$$g_J = 2J + 1$$



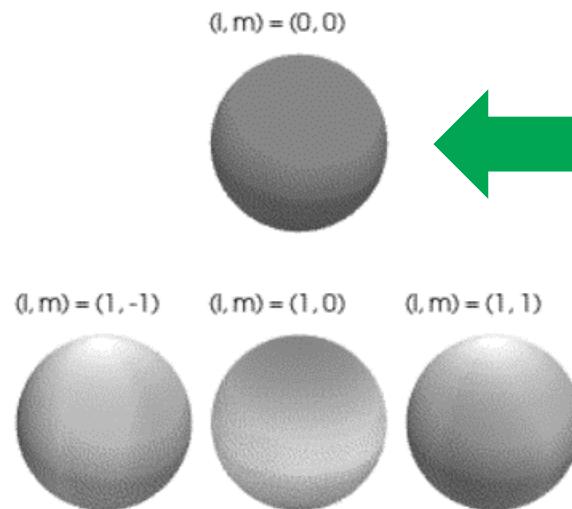
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Uncertainty Principle?

For the rotational ground state, $E_{J=0} = 0$

Doesn't it violate the uncertainty principle?

No!



Model for Molecular Rotation

- Angular velocity ω

$$v_1 = r_1\omega, \quad v_2 = r_2\omega$$

- Kinetic energy

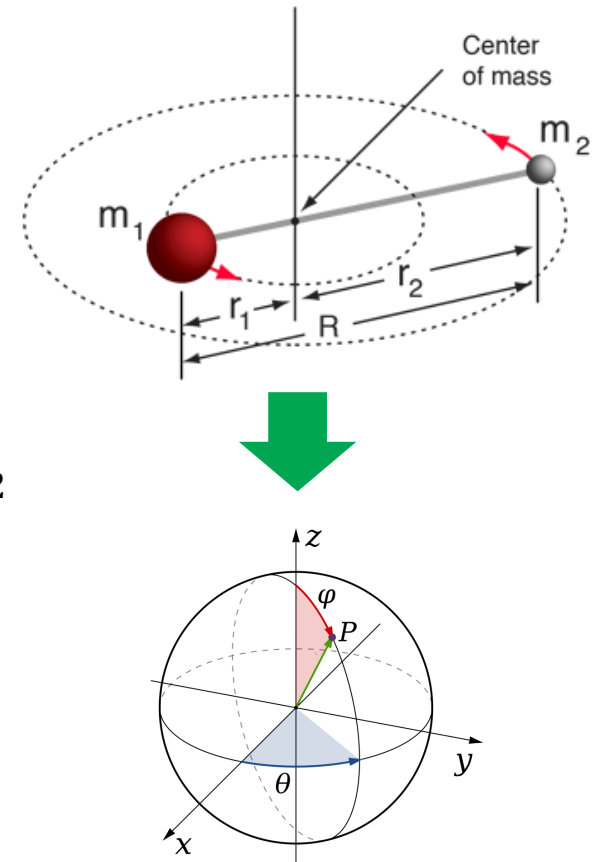
$$E_k = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 = \frac{1}{2}I\omega^2$$

where moment of inertia $I = m_1r_1^2 + m_2r_2^2$

- Angular momentum

$$L = I\omega$$

$$E_k = L^2/2I$$



(Image: <http://hyperphysics.phy-astr.gsu.edu/hbase/molecule/rotrig.html>)

RS 4.5
M 6.7

Selection Rules

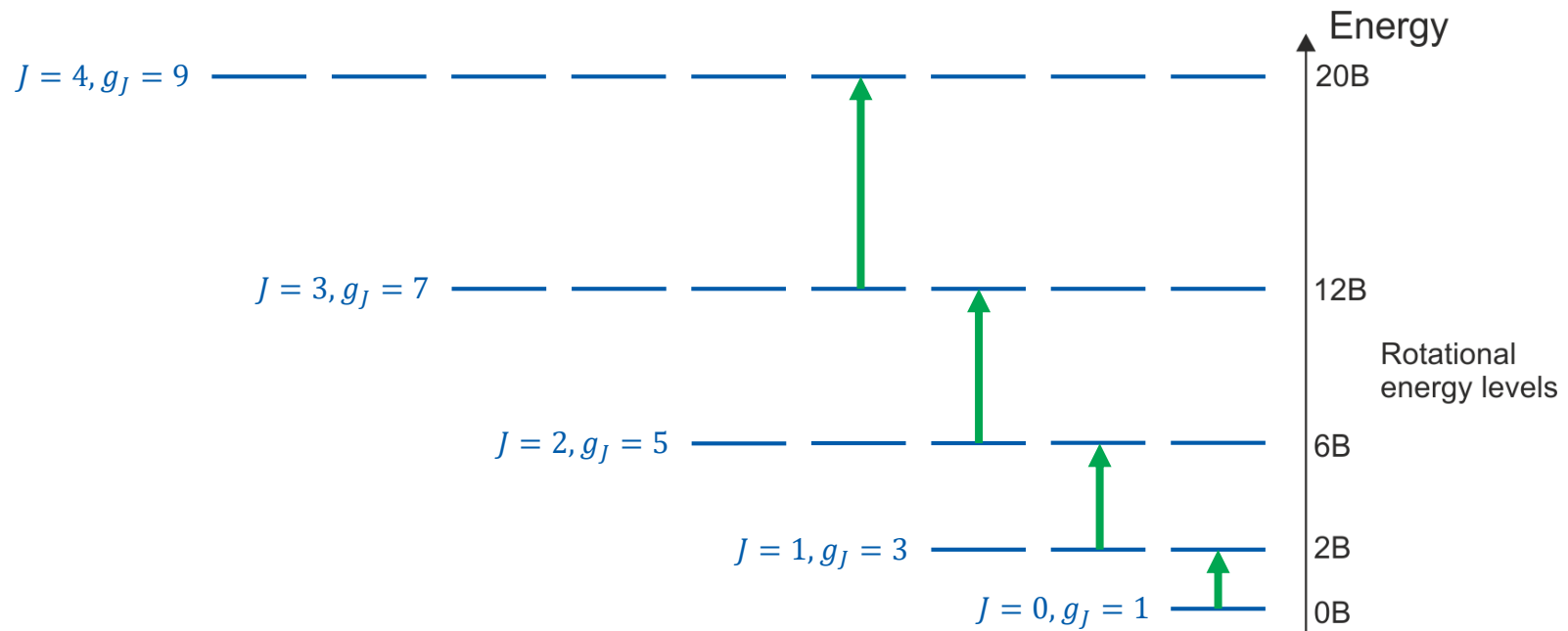
$$\Delta J = \pm 1$$

and

the molecule must possess a permanent dipole moment
(e.g. HCl has a pure rotational spectrum, but N₂ does not)

Energy Levels and Transitions

$$h\nu = \Delta E = (J + 1)(J + 2)B - J(J + 1)B = 2(J + 1)B$$

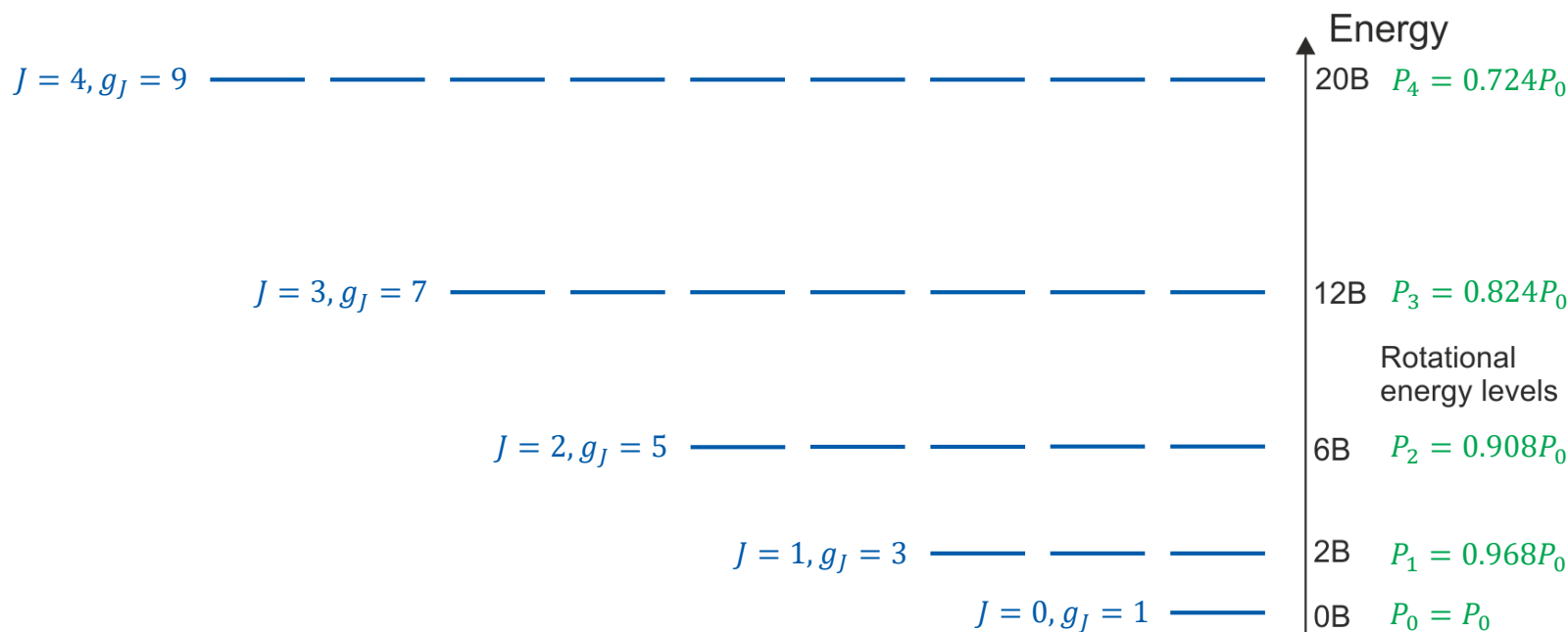


Typical diatomic molecules: $\nu \approx 10^{10}$ - 10^{11} Hz (microwave)

RS 4.5
M 6.7

Populations at RT

$$g_J = 2J + 1$$

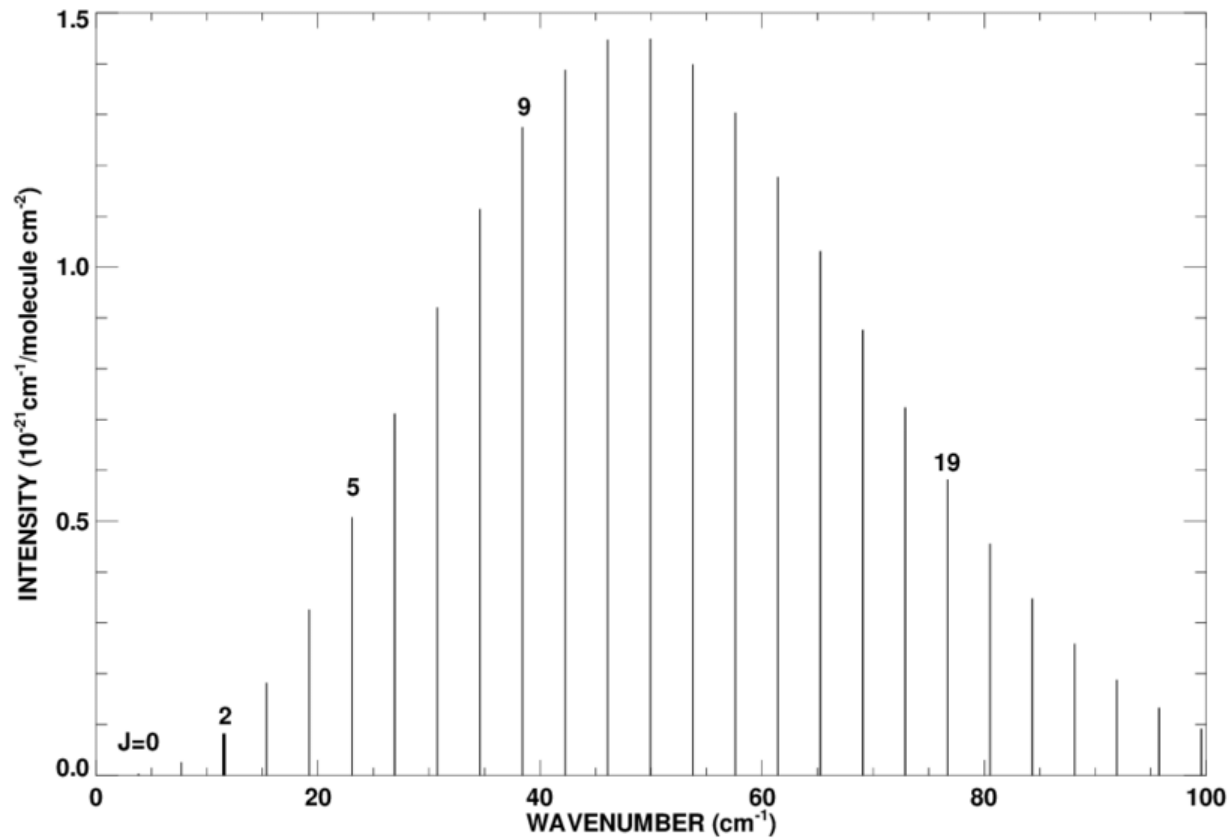


$$P(E_J) \propto e^{-E_J/k_B T} = e^{-BJ(J+1)/k_B T} \sim e^{-0.016J(J+1)} = (0.984)^{J(J+1)}$$

$\uparrow$
 $\nu \sim 10^{11}$ Hz
 $T = 298$ K

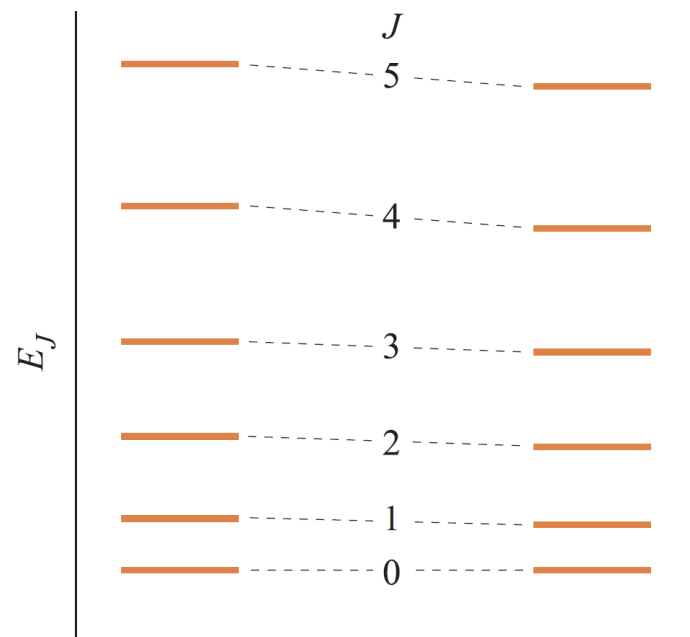
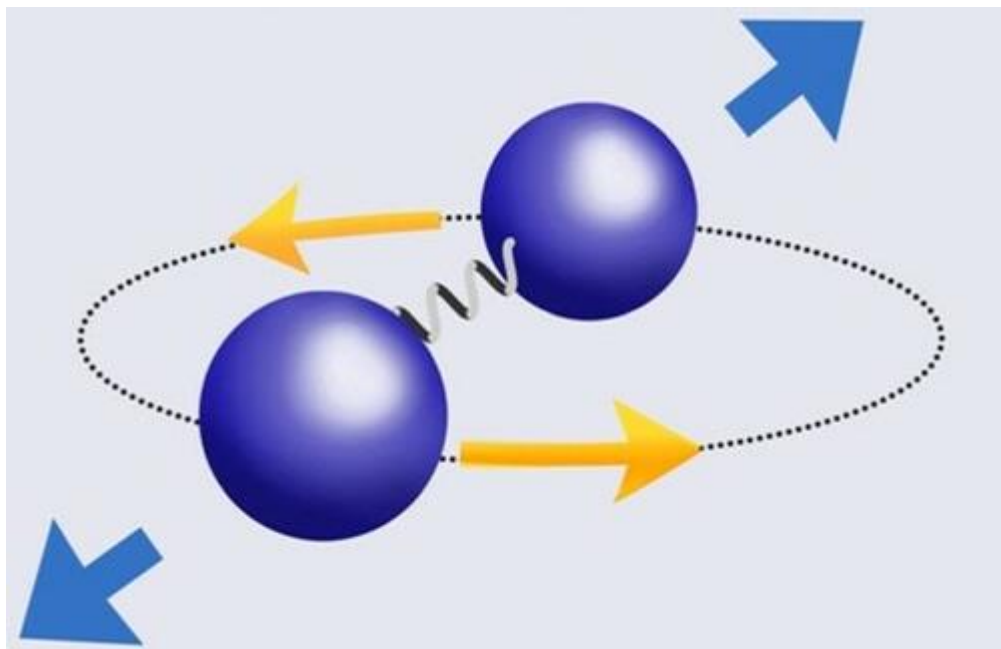


Microwave Spectrum of CO



(Image: https://www.researchgate.net/figure/The-rotational-spectrum-of-CO-The-line-intensities-are-for-a-temperature-of-296-K-The_fig10_253391065)

Centrifugal Distortion



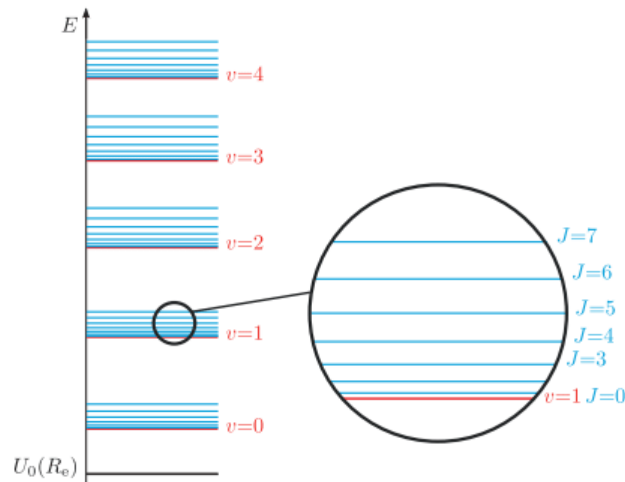
$$E_J = BJ(J + 1) - \underline{D}J^2(J + 1)^2 + \dots$$

centrifugal distortion constant

Rotations and Vibrations

Rigid rotor + harmonic oscillator (no interdependence)

$$E_{v,J} = E_v + E_J = \left(v + \frac{1}{2} \right) h\nu + BJ(J + 1)$$



(Image: https://www.researchgate.net/figure/Vibrational-and-rotational-energy-level-spacing-of-a-diatomic-molecule_fig1_324676654)

Allowed Transitions

- Selection rules: $\Delta v = \pm 1$ and $\Delta J = \pm 1$
- At RT, most vibrational transitions are $0 \rightarrow 1$
- Case 1: $\Delta v = +1$ and $\Delta J = +1$ (R branch)

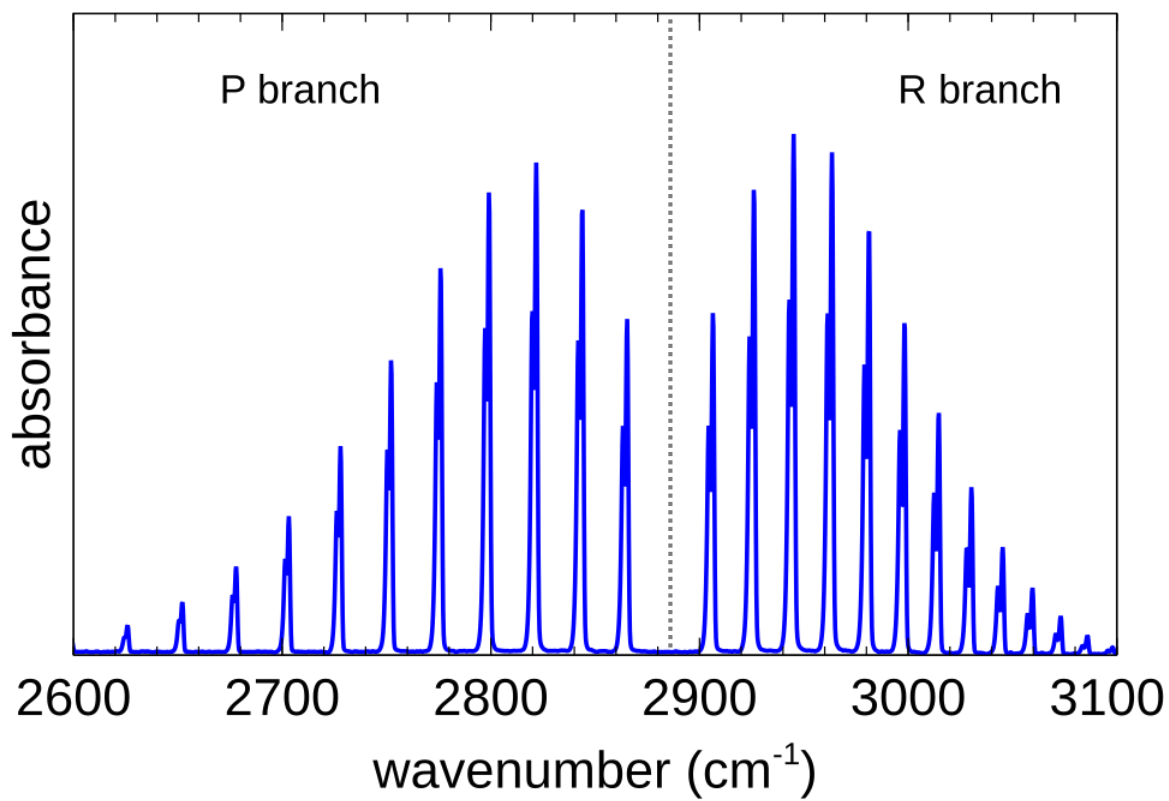
$$\Delta E = E_{v+1,J+1} - E_{v,J} = h\nu + 2B(J + 1)$$

- Case 2: $\Delta v = +1$ and $\Delta J = -1$ (P branch)

$$\Delta E = E_{v+1,J-1} - E_{v,J} = h\nu - 2BJ$$

Rotation-Vibration Spectrum

HCl(g): $0 \rightarrow 1$ vibrational transition



(Image: https://commons.wikimedia.org/wiki/File:Ro-vibrational_spectrum_of_HCl.svg)